About FEMA GRAS Program

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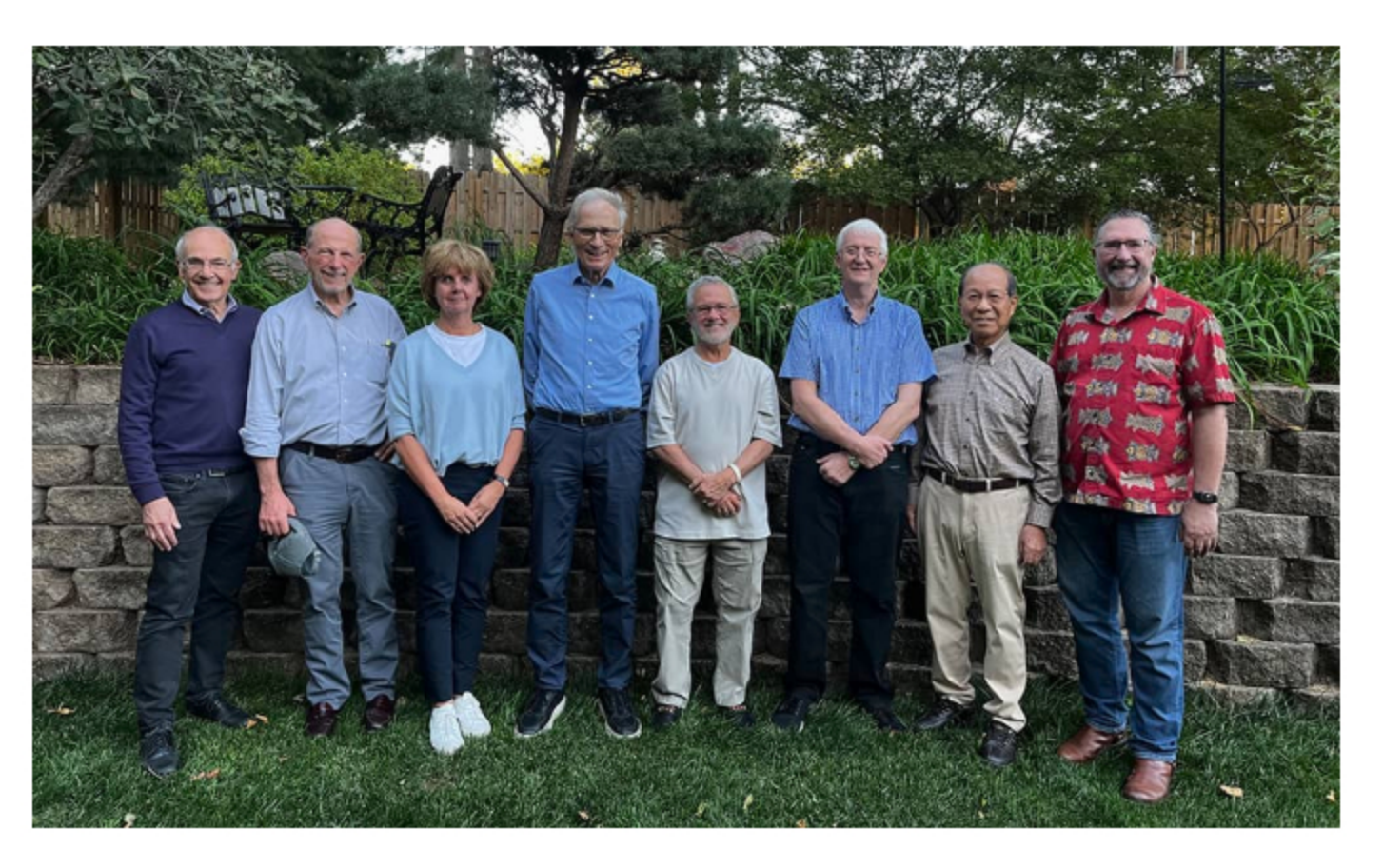

FEMA Expert Panel (left to right): Dr. Steve Hecht, University of Minnesota (Vice-Chair, FEMA Expert Panel); Dr. F. Peter Guengerich, Vanderbilt University; Dr. Ivonne Rietjens, Wageningen University (Emeritus Panel Member); Dr. Gerhard Eisenbrand, University of Kaiserlautern (Retired from faculty position); Dr. Sam Cohen, University of Nebraska Medical Center; Dr. Nigel Gooderham, Imperial College London (Chair, FEMA Expert Panel); Dr. Shoji Fukushima, Osaka City University ; Dr. Thomas J. Rosol, Ohio University.

Introduction

In 1959, The Flavor and Extract Manufacturers Association of the United States (FEMA) took its initial actions to establish a novel program to assess the safety and "GRAS" (generally recognized as safe) status of flavor ingredients as described in the 1958 Food Additives Amendments to the Federal Food, Drug, and Cosmetic Act, the Federal law governing the regulation of flavors and other food ingredients. Since then, the FEMA GRAS program has become the longest-running and most widely recognized industry GRAS assessment program.

The FEMA GRAS program began in 1959 with a survey of the flavor industry to identify flavor ingredients then in use and to provide estimates of the amounts of these substances used to manufacture flavors. The initial FEMA Expert Panel was established in 1960 beginning the Expert Panel's evaluation of the safety of flavor ingredients that continues today. Then and now, the Expert Panel usually consists of six to eight members and over the years has been comprised of top experts in toxicology, organic chemistry, biochemistry, metabolism, and pathology.

The FEMA Expert Panel's GRAS assessments have from the beginning utilized a number of techniques now common in safety evaluations including the use of metabolic studies and structural relationships that had not previously been applied in a significant manner to food ingredients before the Panel did so in the 1960s in its evaluation of flavors. While the Expert Panel has been provided with financial support by FEMA it has always maintained its full independence in its operations and GRAS determinations and follows strict conflict of interest procedures.

The FEMA GRAS program is open to participation by Active Members of FEMA who may submit applications to the Expert Panel requesting consideration of flavor ingredients as generally recognized as safe.

Two publications by Hallagan and Hall (1995, 2009) provide a detailed review of the history of the FEMA GRAS Program, and its legal and scientific bases.

The GRAS Concept

In the United States, the GRAS concept is one way in which the regulatory authority to use a food ingredient (other than color additives) can be determined with the other key path being through an application to the Food and Drug Administration for food additive status. GRAS status may be achieved either through the FDA's voluntary GRAS notification program (FDA, 1997) or through a properly conducted GRAS determination made by a private party. Section 201(s) of the Federal Food, Drug, and Cosmetic Act provides:

The term food additive means any substance the intended use of which results or may reasonably be expected to result, directly or indirectly, in its becoming a component or otherwise affecting the characteristics of any food . . . if such substance is not generally recognized, among experts qualified by training and experience to evaluate its safety as having been adequately shown through scientific procedures (or in the case of a substance used in food prior to January 1, 1958, through either scientific procedures or experience based on common use in food) to be safe under the conditions of intended use . . .

The statutory definition of GRAS has four key criteria, all of which must be met for a food ingredient to be considered generally recognized as safe and exempt from the requirements for food additive approval:

1. There must be general recognition of safety by qualified experts.
2. The experts must be qualified by training and experience to evaluate the substance's safety.
3. The experts must base their determination of safety on scientific procedures or on common use in food prior to 1958.
4. The determination of general recognition of safety must take into account the conditions of intended use for the substance, in other words its function in the food, e.g. flavoring.

A key part of the requirements for GRAS status is that there must be general recognition of safety and not just an assertion of safety *per se*. This means that there must be an opportunity for interested parties to review the basis for GRAS determinations not just the conclusions. Detailed discussions of the requirements for determining GRAS status are available in Hallagan and Hall (1995, 2009) and the many references cited in these two publications.

How Does the **FEMA GRAS** Program Meet the Requirements for GRAS?

Qualified Experts

The mission of the FEMA Expert Panel is to protect the public health through the effective safety evaluation of flavor ingredients using the best available scientific procedures and information.

Over the years, members of the FEMA Expert Panel have included many prominent expert scientists. Most often, members of the FEMA Expert Panel are scientific experts from academia. Members of the Expert Panel have never been employees of FEMA member companies or employees of companies in the food industry. Current members of the FEMA Expert Panel are:

- Ivonne M.C.M. Rietjens, Ph.D. – Chair
  Wageningen University, Netherlands
- Samuel M. Cohen, M.D., Ph.D.
  University of Nebraska
- Dr. Gerhard Eisenbrand, Ph.D. (Retired)
  University of Kaiserlautern
- Shoji Fukushima, M.D., Ph.D.
  Japan Bioassay Research Center
- Nigel J. Gooderham, Ph.D. - Vice Chair
  Imperial College London
- F. Peter Guengerich, Ph.D.
  Vanderbilt University
- Stephen S. Hecht, Ph.D.
  University of Minnesota
- Tom Rosol, D.V.M., Ph.D.
  Ohio University

Scientific Procedures to Determine Safety

As part of the FEMA GRAS program's policy of transparency, the scientific criteria employed by the FEMA Expert Panel for its GRAS assessments are published in the peer-reviewed scientific literature so that they may be reviewed and understood by regulators, business, and the public. The current criteria for the GRAS assessment of single, chemically defined flavoring substances are described by Smith et al. (2005a). The current criteria for natural flavor complexes are described in Smith et al. (2004) and Smith et al. (2005b). The safety standard applied by the FEMA Expert Panel is the same standard required by FDA, a "reasonable certainty of no harm."

Safety Under Conditions of Intended Use

The FEMA Expert Panel only evaluates substances for GRAS status that are used to formulate flavors to be added to human foods. The Expert Panel does not evaluate food ingredients with functions other than flavoring nor does it evaluate flavorings for use in products other than human food. For example, the Expert Panel does not evaluate flavor ingredients for use in tobacco products, e-cigarettes, or other products that involve routes of exposure other than ingestion.

General Recognition of Safety

The general recognition requirement for GRAS status is perhaps the most important factor. The Expert Panel and the FEMA GRAS program meet this requirement in a number of ways:

1. The Expert Panel is generally composed of six to eight members of various scientific disciplines. This provides a high degree of confidence that decisions made by the Panel are consistent with current scientific concepts and that the decisions would be generally recognized as scientifically appropriate.
2. The scientific criteria employed by the FEMA Expert Panel for its GRAS assessments are published in the peer-reviewed scientific literature so that they may be reviewed and understood by regulators, business, and the public. (Smith et al., 2004; Smith et al., 2005; Cohen et al., 2018).
3. The identity of all FEMA GRAS substances is published in several ways. Lists of FEMA GRAS flavor ingredients are published regularly in the widely-read food industry journal, *Food Technology*. The identity of all FEMA GRAS substances is also available on the FEMA website (www.femaflavor.org). This ready availability of information provides the public with notice of FEMA GRAS determinations allowing for challenges to the Expert Panel's conclusions.
4. The scientific information serving as the basis for the Expert Panel's determinations of GRAS status is provided to FDA for all FEMA GRAS flavor ingredients. This allows the agency to include the information in its databases and to challenge any GRAS determinations that it wishes. It is important to note that the receipt by FDA of the information provided by FEMA does not constitute FDA's "approval" of the GRAS determinations made by the FEMA Expert Panel.
5. Descriptions of the scientific information supporting the GRAS status of individual flavor ingredients are published in the "FEMA group summaries" in which information on individual flavoring substances and their structural relatives is reviewed. These summaries are published in the widely-read toxicology journal *Food and Chemical Toxicology*.
6. Group summaries on approximately 2,200 FEMA GRAS flavoring substances have also been published by the Joint FAO/WHO Expert Committee on Food Additives (JECFA), an international scientific advisory body managed through the United Nations (See http://www.who.int/foodsafety/areas_work/chemical-risks/jecfa/en/).
7. Scientific Literature Reviews (SLRs) published by the National Technical Information Service (NTIS), a federal government agency, provide a large amount of information relevant to the safety of FEMA GRAS flavor ingredients up through the late 1980s.
8. FEMA publishes detailed reviews on the safety of individual flavoring substances to address specific issues.

In addition to the multiple sources of published information listed above, FEMA's policy is to share information on all FEMA GRAS substances with anyone upon request. For example, if a person wants to question or challenge the status of a FEMA GRAS substance the person may obtain from FEMA all of the available information used to determine that the substance is GRAS under its conditions of intended use as a flavor ingredient.

Additional support to meet the general recognition requirement is FDA's acknowledgment in *Federal Register* notices that FEMA GRAS substances were "approved for inclusion in the FDA GRAS review process" and therefore were eligible for the application of its bulk flavor labeling policy. (41 *Fed. Reg.* 4954. 3 February 1976). FDA's 1976 notice explicitly accounted for "FEMA GRAS lists 3 through 9." (41 *Fed. Reg.* 4954. 3 February 1976). Subsequent notices accounted for new FEMA GRAS substances as follows:

- "GRAS 10" – 42 *Fed. Reg.* 55643. 18 October 1977
- "GRAS 11" – 43 *Fed. Reg.* 22784. 26 May 1978
- "GRAS 12" – 44 *Fed. Reg.* 71460. 11 December 1979

FDA published its last notice in 1979 due to a change in agency priorities.

FEMA Expert Panel Conflict of Interest Protections and Procedures

A number of measures are in place to assure that Expert Panel decisions on GRAS status are fully objective and based solely on the merits of the available information and are not subject to bias.

1. FEMA Expert Panel members are not allowed to have consulting relationships with FEMA member companies regarding anything to do with flavors in the context of the FEMA GRAS Program.
2. FEMA Expert Panel members provide a declaration of their consulting and business relationships prior to each Expert Panel meeting for review and action, if necessary, by the Panel's Legal Advisor. Action may include mandated recusal at meetings.
3. FEMA Expert Panel members do not prepare GRAS applications. Applications for FEMA GRAS assessments are prepared by the FEMA member seeking the GRAS assessment for the substance at issue.
4. When evaluating GRAS applications, Expert Panel members do not know the identity of the company responsible for the application. The identity of applicant-companies is maintained as confidential information by the FEMA staff even after the substance is evaluated and granted GRAS status.
5. Companies submitting GRAS applications are not allowed to contact Expert Panel members in any way nor are they allowed to attend meetings during which their applications are being considered.
6. The identity of the members of the Expert Panel is known to the public through their regular publications in *Food Technology*, *Food and Chemical Toxicology*, and other publicly available journals.
7. The Expert Panel is self-appointed – members are not appointed by FEMA. When a member retires from the Panel, the retiring member recommends a successor whose qualifications are reviewed by the remaining Expert Panel members who make the appointment.
8. FEMA Expert Panel members conduct their reviews of GRAS applications during in-person meetings, usually three times per year and receive an honorarium for their service whether or not they conclude that any substances are GRAS.
9. FEMA staff members are not allowed to have independent consulting or business relationships with FEMA member companies regarding anything to do with flavors in the context of the FEMA GRAS program.

The FEMA GRAS Program and Transparency

The FEMA GRAS program meets the highest standards for transparency. All GRAS determinations made by the FEMA Expert Panel are provided to FDA along with the scientific information supporting the determinations. It is important to note that the receipt by FDA of the information provided by FEMA does not constitute FDA's "approval" of the GRAS determinations made by the FEMA Expert Panel. The identities of all flavor ingredients determined to be GRAS are published regularly in the widely read journal *Food Technology*. The scientific basis for the Expert Panel's GRAS determinations including detailed summaries of the supporting scientific information are published in widely read scientific journals including *Food and Chemical Toxicology*. Extensive summaries of the information supporting the safety of FEMA GRAS substances are also published by the Joint FAO/WHO Expert Committee on Food Additives (JECFA), an independent international scientific advisory body managed through the United Nations (See http://apps.who.int/food-additives-contaminants-jecfa-database/search.aspx).

A significant additional transparency measure is that anyone wishing to review the scientific information supporting the GRAS status of any FEMA GRAS flavor ingredient may request the information from FEMA and receive it for only the cost of duplication.

Evidence of the high degree of transparency of the FEMA GRAS program can be found in a list of more than two hundred individual publications describing the FEMA GRAS program and the safety of flavor ingredients. This list is available upon request from FEMA.

In conclusion, the FEMA GRAS program meets the highest standards for transparency and in doing so provides a high level of assurance in its GRAS determinations for flavor ingredients.

For more information on the FEMA GRAS program please contact John Hallagan – FEMA Senior Advisor and General Counsel (Hondobear@aol.com) or Sean Taylor – Scientific Secretary to the FEMA Expert Panel (staylor@vertosolutions.net).

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